

```

from pyspark.sql import SparkSession
from pyspark.sql.functions import col

spark = SparkSession.builder.appName("Week6").getOrCreate()

df = spark.read \
    .option("header", True) \
    .option("inferSchema", True) \
    .option("multiLine", True) \
    .option("escape", "\\") \
    .csv("Sample - Superstore.csv")

df.printSchema()

root
|-- Row ID: integer (nullable = true)
|-- Order ID: string (nullable = true)
|-- Order Date: string (nullable = true)
|-- Ship Date: string (nullable = true)
|-- Ship Mode: string (nullable = true)
|-- Customer ID: string (nullable = true)
|-- Customer Name: string (nullable = true)
|-- Segment: string (nullable = true)
|-- Country: string (nullable = true)
|-- City: string (nullable = true)
|-- State: string (nullable = true)
|-- Postal Code: integer (nullable = true)
|-- Region: string (nullable = true)
|-- Product ID: string (nullable = true)
|-- Category: string (nullable = true)
|-- Sub-Category: string (nullable = true)
|-- Product Name: string (nullable = true)
|-- Sales: double (nullable = true)
|-- Quantity: integer (nullable = true)
|-- Discount: double (nullable = true)
|-- Profit: double (nullable = true)

print(df.count())

9994

df.select("Sales").show(20, truncate=False)

```

```

+-----+
|Sales  |
+-----+
|261.96 |
|731.94 |
|14.62  |
|957.5775|

```

```
|22.368 |
|48.86  |
|7.28   |
|907.152|
|18.504 |
|114.9   |
|1706.184|
|911.424 |
|15.552 |
|407.976 |
|68.81  |
|2.544  |
|665.88 |
|55.5   |
|8.56   |
|213.48 |
```

```
+-----+
```

only showing top 20 rows

```
df.select("Product Name").show(20, truncate=False)
```

```
+-----+
```

```
-----+
```

```
|Product Name
```

```
|
```

```
+-----+
```

```
-----+
```

```
|Bush Somerset Collection Bookcase
```

```
|
```

```
|Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back
```

```
|
```

```
|Self-Adhesive Address Labels for Typewriters by Universal
```

```
|
```

```
|Bretford CR4500 Series Slim Rectangular Table
```

```
|
```

```
|Eldon Fold 'N Roll Cart System
```

```
|
```

```
|Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood
```

```
|
```

```
|Newell 322
```

```
|
```

```
|Mitel 5320 IP Phone VoIP phone
```

```
|
```

```
|DXL Angle-View Binders with Locking Rings by Samsill
```

```
|
```

```
|Belkin F5C206VTEL 6 Outlet Surge
```

```
|
```

```
|Chromcraft Rectangular Conference Tables
```

```
|
```

```
|Konftel 250 Conferencephone- Charcoal black
```

```

|Xerox 1967
|Fellowes PB200 Plastic Comb Binding Machine
|Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA
Filter|
|Storex DuraTech Recycled Plastic Frosted Binders
|
|Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D
|
|Fellowes Super Stor/Drawer
|
|Newell 341
|
|Cisco SPA 501G IP Phone
|
+-----+
-----+
only showing top 20 rows

df = df.withColumnRenamed(
    "Sales",
    "Total_Sales"
)

df.printSchema()

root
|-- Row ID: integer (nullable = true)
|-- Order ID: string (nullable = true)
|-- Order Date: string (nullable = true)
|-- Ship Date: string (nullable = true)
|-- Ship Mode: string (nullable = true)
|-- Customer ID: string (nullable = true)
|-- Customer Name: string (nullable = true)
|-- Segment: string (nullable = true)
|-- Country: string (nullable = true)
|-- City: string (nullable = true)
|-- State: string (nullable = true)
|-- Postal Code: integer (nullable = true)
|-- Region: string (nullable = true)
|-- Product ID: string (nullable = true)
|-- Category: string (nullable = true)
|-- Sub-Category: string (nullable = true)
|-- Product Name: string (nullable = true)
|-- Total_Sales: double (nullable = true)
|-- Quantity: integer (nullable = true)
|-- Discount: double (nullable = true)

```

```
|-- Profit: double (nullable = true)
```

```
df = df.withColumn(  
    "Final_Price",  
    col("Total_Sales") * 1.18  
)
```

```
df.select(  
    "Total_Sales",  
    "Final_Price"  
)>.show(10)
```

```
+-----+-----+  
|Total_Sales|      Final_Price|  
+-----+-----+  
|      261.96|309.11279999999994|  
|      731.94|      863.6892|  
|       14.62|      17.2516|  
|    957.5775|    1129.94145|  
|      22.368|26.394239999999996|  
|      48.86|57.654799999999994|  
|       7.28|      8.5904|  
|    907.152|   1070.43936|  
|     18.504|     21.83472|  
|     114.9|     135.582|  
+-----+-----+
```

only showing top 10 rows

```
df.select(  
    "Customer ID",  
    "Product Name",  
    "Total_Sales"  
)>.show(10)
```

```
+-----+-----+-----+  
|Customer ID|      Product Name|Total_Sales|  
+-----+-----+-----+  
|    CG-12520|Bush Somerset Col...|      261.96|  
|    CG-12520|Hon Deluxe Fabric...|      731.94|  
|    DV-13045|Self-Adhesive Add...|       14.62|  
|    SO-20335|Bretford CR4500 S...|    957.5775|  
|    SO-20335|Eldon Fold 'N Rol...|     22.368|  
|    BH-11710|Eldon Expressions...|     48.86|  
|    BH-11710|      Newell 322|       7.28|  
|    BH-11710|Mitel 5320 IP Pho...|    907.152|  
|    BH-11710|DXL Angle-View Bi...|     18.504|  
|    BH-11710|Belkin F5C206VTEL...|     114.9|  
+-----+-----+-----+
```

only showing top 10 rows

```
df.filter(
    col("Category") == "Technology"
).show(10)
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|Row ID|      Order ID|Order Date| Ship Date|      Ship Mode|Customer
ID|      Customer Name|      Segment|      Country|      City|
State|Postal Code| Region|      Product ID|  Category|Sub-Category|
Product Name|Total_Sales|Quantity|Discount|  Profit|
Final_Price|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|      8|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class|  BH-
11710|  Brosina Hoffman|  Consumer|United States|  Los Angeles|
California|      90032|  West|TEC-PH-10002275|Technology|
Phones|Mitel 5320 IP Pho...|      907.152|      6|      0.2| 90.7152|
1070.43936|
|     12|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class|  BH-
11710|  Brosina Hoffman|  Consumer|United States|  Los Angeles|
California|      90032|  West|TEC-PH-10002033|Technology|
Phones|Konftel 250 Confe...|      911.424|      4|      0.2| 68.3568|
1075.48032|
|     20|CA-2014-143336| 8/27/2014| 9/1/2014|  Second Class|  ZD-
21925|Zuschuss Donatelli|  Consumer|United States|San Francisco|
California|      94109|  West|TEC-PH-10001949|Technology|
Phones|Cisco SPA 501G IP...|      213.48|      3|      0.2| 16.011|
251.90639999999999|
|     27|CA-2016-121755| 1/16/2016| 1/20/2016|  Second Class|  EH-
13945|  Eric Hoffmann|  Consumer|United States|  Los Angeles|
California|      90049|  West|TEC-AC-10003027|Technology|
Accessories|Imation 8GB Mini ...|      90.57|      3|      0.0|
11.7741|106.87259999999999|
|     36|CA-2016-117590| 12/8/2016|12/10/2016|  First Class|  GH-
14485|      Gene Hale|  Corporate|United States|  Richardson|
Texas|      75080|Central|TEC-PH-10004977|Technology|  Phones|
GE 30524EE4|      1097.544|      7|      0.2|123.4737|      1295.10192|
|     41|CA-2015-117415|12/27/2015|12/31/2015|Standard Class|  SN-
20710|      Steve Nguyen|Home Office|United States|      Houston|
Texas|      77041|Central|TEC-PH-10000486|Technology|  Phones|
Plantronics HL10 ...|      371.168|      4|      0.2| 41.7564|
437.97823999999997|
|     42|CA-2017-120999| 9/10/2017| 9/15/2017|Standard Class|  LC-
16930|      Linda Cazamias|  Corporate|United States|  Naperville|
```

```

Illinois|      60540|Central|TEC-PH-10004093|Technology|      Phones|
Panasonic Kx-TS550|      147.168|      4|      0.2| 16.5564|
173.65824|
|      45|CA-2016-118255| 3/11/2016| 3/13/2016|      First Class|      ON-
18715|      Odella Nelson|      Corporate|United States|      Eagan|
Minnesota|      55122|Central|TEC-AC-10000171|Technology| Accessories|
Verbatim 25 GB 6x...|      45.98|      2|      0.0| 19.7714|
54.256399999999999|
|      48|CA-2016-169194| 6/20/2016| 6/25/2016|Standard Class|      LH-
16900|      Lena Hernandez|      Consumer|United States|      Dover|
Delaware|      19901|      East|TEC-AC-10002167|Technology| Accessories|
Imation 8gb Micro...|      45.0|      3|      0.0| 4.95|
53.0999999999999994|
|      49|CA-2016-169194| 6/20/2016| 6/25/2016|Standard Class|      LH-
16900|      Lena Hernandez|      Consumer|United States|      Dover|
Delaware|      19901|      East|TEC-PH-10003988|Technology|      Phones|
LF Elite 3D Dazzl...|      21.8|      2|      0.0| 6.104|
25.724|

```

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+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
only showing top 10 rows

```

```

df.filter(
    (col("Region") == "West") &
    (col("Total_Sales") > 100)
).show()

```

```

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+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|Row ID|      Order ID|Order Date| Ship Date|      Ship Mode|Customer
ID|      Customer Name|      Segment|      Country|      City|
State|Postal Code|Region|      Product ID|      Category|Sub-Category|
Product Name|Total_Sales|Quantity|Discount|      Profit|
Final_Price|
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|      8|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class|      BH-
11710|      Brosina Hoffman|      Consumer|United States|      Los Angeles|
California|      90032|      West|TEC-PH-10002275|      Technology|
Phones|Mitel 5320 IP Pho...|      907.152|      6|      0.2| 90.7152|
1070.43936|

```

| 10|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-11710| Brosina Hoffman| Consumer|United States| Los Angeles| California| 90032| West|OFF-AP-10002892|Office Supplies| Appliances|Belkin F5C206VTEL...| 114.9| 5| 0.0| 34.47| 135.582|

| 11|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-11710| Brosina Hoffman| Consumer|United States| Los Angeles| California| 90032| West|FUR-TA-10001539| Furniture| Tables|Chromcraft Rectan...| 1706.184| 9| 0.2| 85.3092| 2013.29712|

| 12|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class| BH-11710| Brosina Hoffman| Consumer|United States| Los Angeles| California| 90032| West|TEC-PH-10002033| Technology| Phones|Konftel 250 Confe...| 911.424| 4| 0.2| 68.3568| 1075.48032|

| 14|CA-2016-161389| 12/5/2016|12/10/2016|Standard Class| IM-15070| Irene Maddox| Consumer|United States| Seattle| Washington| 98103| West|OFF-BI-10003656|Office Supplies| Binders|Fellowes PB200 Pl...| 407.976| 3| 0.2|132.5922| 481.41168|

| 20|CA-2014-143336| 8/27/2014| 9/1/2014| Second Class| ZD-21925|Zuschuss Donatelli| Consumer|United States|San Francisco| California| 94109| West|TEC-PH-10001949| Technology| Phones|Cisco SPA 501G IP...| 213.48| 3| 0.2| 16.011| 251.90639999999996|

| 25|CA-2015-106320| 9/25/2015| 9/30/2015|Standard Class| EB-13870| Emily Burns| Consumer|United States| Orem| Utah| 84057| West|FUR-TA-10000577| Furniture| Tables| Bretford CR4500 S...| 1044.63| 3| 0.0|240.2649| 1232.6634000000001|

| 65|CA-2015-135545|11/24/2015|11/30/2015|Standard Class| KM-16720| Kunst Miller| Consumer|United States| Los Angeles| California| 90004| West|OFF-PA-10003892|Office Supplies| Paper| Xerox 1943| 146.73| 3| 0.0| 68.9631| 173.14139999999998|

| 68|CA-2014-106376| 12/5/2014|12/10/2014|Standard Class| BS-11590| Brendan Sweed| Corporate|United States| Gilbert| Arizona| 85234| West|OFF-AR-10002671|Office Supplies| Art|Hunt BOSTON Model...| 1113.024| 8| 0.2|111.3024| 1313.3683199999998|

| 69|CA-2014-106376| 12/5/2014|12/10/2014|Standard Class| BS-11590| Brendan Sweed| Corporate|United States| Gilbert| Arizona| 85234| West|TEC-PH-10002726| Technology| Phones|netTALK DUO VoIP ...| 167.968| 4| 0.2| 62.988| 198.20224|

| 104|US-2015-156867|11/13/2015|11/17/2015|Standard Class| LC-16870| Lena Cacioppo| Consumer|United States| Aurora| Colorado| 80013| West|TEC-AC-10001552| Technology| Accessories|Logitech K350 2.4...| 238.896| 6| 0.2|-

26.8758|281.89727999999997|
 | 105|US-2015-156867|11/13/2015|11/17/2015|Standard Class| LC-
 16870| Lena Cacioppo| Consumer|United States| Aurora|
 Colorado| 80013| West|FUR-FU-10004006| Furniture|
 Furnishings|Deflect-o DuraMat...| 102.36| 3| 0.2| -
 3.8385|120.78479999999999|
 | 118|CA-2015-110457| 3/2/2015| 3/6/2015|Standard Class| DK-
 13090| Dave Kipp| Consumer|United States| Seattle|
 Washington| 98103| West|FUR-TA-10001768| Furniture|
 Tables|Hon Racetrack Con...| 787.53| 3| 0.0|165.3813|
 929.28539999999999|
 | 128|US-2017-107272| 11/5/2017|11/12/2017|Standard Class| TS-
 21610| Troy Staebel| Consumer|United States| Phoenix|
 Arizona| 85023| West|OFF-ST-10002974|Office Supplies|
 Storage|Trav-L-File Heavy...| 243.992| 7| 0.2| 30.499|
 287.91056|
 | 130|US-2016-125969| 11/6/2016|11/10/2016| Second Class| LS-
 16975| Lindsay Shagiari|Home Office|United States| Los Angeles|
 California| 90004| West|FUR-FU-10003773| Furniture|
 Furnishings|Eldon Cleatmat Pl...| 238.56| 3| 0.0|
 26.2416|281.50079999999997|
 | 144|CA-2017-106180| 9/18/2017| 9/23/2017|Standard Class| SH-
 19975| Sally Hughsby| Corporate|United States|San Francisco|
 California| 94122| West|OFF-PA-10004327|Office Supplies|
 Paper| Xerox 1911| 143.7| 3| 0.0| 68.976|
 169.56599999999997|
 | 146|CA-2015-110744| 9/7/2015| 9/12/2015|Standard Class| HA-
 14920| Helen Andreada| Consumer|United States| Pasadena|
 California| 91104| West|OFF-ST-10003656|Office Supplies|
 Storage|Safco Industrial ...| 671.93| 7| 0.0| 20.1579|
 792.87739999999999|
 | 152|CA-2016-158834| 3/13/2016| 3/16/2016| First Class| TW-
 21025| Tamara Willingham|Home Office|United States| Scottsdale|
 Arizona| 85254| West|OFF-AP-10000326|Office Supplies|
 Appliances|Belkin 7 Outlet S...| 157.92| 5| 0.2|
 17.766|186.34559999999996|
 | 153|CA-2016-158834| 3/13/2016| 3/16/2016| First Class| TW-
 21025| Tamara Willingham|Home Office|United States| Scottsdale|
 Arizona| 85254| West|TEC-PH-10001254| Technology|
 Phones|Jabra BIZ 2300 Du...| 203.184| 2| 0.2| 15.2388|
 239.75712|
 | 155|CA-2015-124919| 5/31/2015| 6/2/2015| First Class| SP-
 20650| Stephanie Phelps| Corporate|United States| San Jose|
 California| 95123| West|OFF-PA-10002254|Office Supplies|
 Paper| Xerox 1883| 105.52| 4| 0.0| 48.5392|
 124.51359999999998|
 +-----+-----+-----+-----+-----+-----+
 +-----+-----+-----+-----+-----+-----+
 +-----+-----+-----+-----+-----+-----+

```
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+-----+-----+-----+-----+
only showing top 20 rows
```

```
df.filter(
    col("Customer ID").isNotNull()
).show(10)
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|Row ID|      Order ID|Order Date| Ship Date|      Ship Mode|Customer
ID|  Customer Name|  Segment|      Country|      City|      State|
Postal Code|Region|      Product ID|      Category|Sub-Category|
Product Name|Total_Sales|Quantity|Discount|  Profit|
Final_Price|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|      1|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|  CG-
12520|  Claire Gute| Consumer|United States|      Henderson|
Kentucky|      42420| South|FUR-B0-10001798|      Furniture|
Bookcases|Bush Somerset Col...|      261.96|      2|      0.0| 41.9136|
309.112799999999994|
|      2|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|  CG-
12520|  Claire Gute| Consumer|United States|      Henderson|
Kentucky|      42420| South|FUR-CH-10000454|      Furniture|
Chairs|Hon Deluxe Fabric...|      731.94|      3|      0.0| 219.582|
863.6892|
|      3|CA-2016-138688| 6/12/2016| 6/16/2016|  Second Class|  DV-
13045|Darrin Van Huff|Corporate|United States|      Los Angeles|
California|      90036|  West|OFF-LA-10000240|Office Supplies|
Labels|Self-Adhesive Add...|      14.62|      2|      0.0| 6.8714|
17.2516|
|      4|US-2015-108966|10/11/2015|10/18/2015|Standard Class|  SO-
20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|
Florida|      33311| South|FUR-TA-10000577|      Furniture|
Tables|Bretford CR4500 S...|      957.5775|      5|      0.45|-383.031|
1129.94145|
|      5|US-2015-108966|10/11/2015|10/18/2015|Standard Class|  SO-
20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|
Florida|      33311| South|OFF-ST-10000760|Office Supplies|
Storage|Eldon Fold 'N Rol...|      22.368|      2|      0.2| 2.5164|
26.394239999999996|
|      6|CA-2014-115812| 6/9/2014| 6/14/2014|Standard Class|  BH-
11710|Brosina Hoffman| Consumer|United States|      Los Angeles|
```

```

California|      90032|  West|FUR-FU-10001487|      Furniture|
Furnishings|Eldon Expressions...|      48.86|      7|      0.0|
14.1694|57.6547999999999994|
|      7|CA-2014-115812|  6/9/2014|  6/14/2014|Standard Class|  BH-
11710|Brosina Hoffman| Consumer|United States|      Los Angeles|
California|      90032|  West|OFF-AR-10002833|Office Supplies|
Art|      Newell 322|      7.28|      4|      0.0|  1.9656|
8.5904|
|      8|CA-2014-115812|  6/9/2014|  6/14/2014|Standard Class|  BH-
11710|Brosina Hoffman| Consumer|United States|      Los Angeles|
California|      90032|  West|TEC-PH-10002275|      Technology|
Phones|Mitel 5320 IP Pho...|      907.152|      6|      0.2| 90.7152|
1070.43936|
|      9|CA-2014-115812|  6/9/2014|  6/14/2014|Standard Class|  BH-
11710|Brosina Hoffman| Consumer|United States|      Los Angeles|
California|      90032|  West|OFF-BI-10003910|Office Supplies|
Binders|DXL Angle-View Bi...|      18.504|      3|      0.2|  5.7825|
21.83472|
|     10|CA-2014-115812|  6/9/2014|  6/14/2014|Standard Class|  BH-
11710|Brosina Hoffman| Consumer|United States|      Los Angeles|
California|      90032|  West|OFF-AP-10002892|Office Supplies|
Appliances|Belkin F5C206VTEL...|      114.9|      5|      0.0|
34.47|      135.582|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
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only showing top 10 rows

df.write \
.mode("overwrite") \
.option("header", "true") \
.csv("Week6_Output")

parquet_df = spark.read.parquet(
    "Week6_Parquet"
)

parquet_df.show(5)

spark.stop()

```

Q1. Explain the roles of the Driver, Cluster Manager, and Executor in a Spark application.

Answer:

Driver: Controls the Spark application, creates the SparkSession, schedules tasks, and collects results. Cluster Manager: Allocates resources (CPU and memory) for Spark applications. Examples include Standalone, YARN, Mesos, and Kubernetes. Executor: Executes tasks assigned by the Driver, stores intermediate data, and returns results.

Q2. How does Spark's Lazy Evaluation strategy improve performance when chain-processing large datasets?

Answer:

Spark records all transformations but does not execute them immediately. Execution starts only when an action is called. This allows Spark to optimize the execution plan using a Directed Acyclic Graph (DAG), reducing unnecessary computations and improving performance.

Q3. Read a CSV file. `df = spark.read .option("header", "true") .option("inferSchema", "true") .csv("data/source.csv")`

Q4. Difference between CSV and Parquet. CSV Parquet Row-based storage Columnar storage Larger file size Compressed storage Slower query performance Faster query performance No schema support Stores schema information

Q5. Select product_id and price where category is Electronics. `df.filter( df["category"] == "Electronics" ).select( "product_id", "price" ).show()`

Q6. Rename a column and cast data type. `from pyspark.sql.functions import col`

```
df = df.withColumnRenamed( "old_name", "new_name" )
```

```
df = df.withColumn( "price", col("price").cast("double") )
```

Q7. Explain Lineage Graph (DAG).

Answer:

Spark maintains a Lineage Graph (DAG) that records all transformations. If a worker node fails, Spark recomputes only the lost partitions instead of the entire dataset, providing fault tolerance.

Q8. Filter completed orders with amount greater than 1000.

```
df_orders.filter( (df_orders["status"] == "Completed") & (df_orders["amount"] > 1000) ).show()
```

Q9. Explain Predicate Pushdown.

Answer:

Predicate Pushdown is an optimization where Spark sends filter conditions directly to the Parquet reader. Only matching rows are read from disk, reducing memory usage and improving performance.

Q10. Add a new column. `from pyspark.sql.functions import col`

```
df = df.withColumn( "final_price", col("base_price") * 1.18 )
```

Q11. What is the difference between Transformations and Actions? Give two examples of each.

Answer:

In Apache Spark, Transformations and Actions are two types of operations performed on DataFrames or RDDs.

Transformations create a new DataFrame or RDD from an existing one without executing the operation immediately. Spark records these operations and executes them only when an action is called (Lazy Evaluation). Actions trigger the execution of all pending transformations and either return a result to the Driver or write data to storage.

Q12. Load Parquet, remove null values, and save as CSV. `df = spark.read.parquet("path/to/input")`

`df.filter( df["user_id"].isNotNull() ).write .mode("overwrite") .option("header", "true") .csv("path/to/output")`

Q13. Difference between Client Mode and Cluster Mode. Client Mode Cluster Mode Driver runs on client machine Driver runs on cluster Used for development Used for production Client must stay connected Client can disconnect after submission

Q14. Filter rows where region is North OR priority is High. `df.filter( (df["region"] == "North") | (df["priority"] == "High") ).show()`

Q15. Why use `.show(5)` instead of `.collect()`?

Answer:

`.show(5)` displays only the first five rows, making it memory-efficient. `.collect()` loads the entire dataset into the Driver's memory, which can cause memory overflow or application failure for very large datasets.